

HW3

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```
# Assuming b = X and c = Y are known values based on the actual data
# Construct the 2x2 contingency table
library(tinytex)
library(readxl)
data3 <- read_excel("q3_data.xlsx")
head(data3)
```

```
# A tibble: 6 x 3
      ID `Before the debate` `After the debate`
  <dbl> <chr>                <chr>
1     1 A                  A
2     2 A                  A
3     3 A                  A
4     4 A                  A
5     5 A                  A
6     6 A                  A
```

```
# Step 2: Create the 2x2 contingency table
table_data <- table(data3$`Before the debate`, data3$`After the debate`)
print(table_data)
```

```
      A B
A 12  3
B  5 30
```

```
# It should look like:
```

```
#      After
# Before  A   B
#      A  xx yy
#      B  zz ww
```

```
library(Hmisc)
```

Attaching package: 'Hmisc'

The following objects are masked from 'package:base':

```
format.pval, units
```

```
latex(describe(data3), file = "", caption.placement = "top")
```

	data3
3 Variables	50 Observations

```
ID
n missing distinct Info Mean Gmd .05 .10 .25 .50 .75 .90 .95
50      0      50     1  25.5  17  3.45 5.90 13.25 25.50 37.75 45.10 47.55

lowest :  1  2  3  4  5, highest: 46 47 48 49 50
```

Before the debate

```
n missing distinct
50      0      2

Value      A  B
Frequency  15 35
Proportion 0.3 0.7
```

After the debate

```
n missing distinct
50      0      2

Value      A  B
Frequency  17 33
Proportion 0.34 0.66
```

```
# Perform McNemar's test
result <- mcnemar.test(table_data,correct = FALSE)
print(result)
```

McNemar's Chi-squared test

```
data:  table_data
McNemar's chi-squared = 0.5, df = 1, p-value = 0.4795
```

從McNemar's Chi-squared test中可以得知其結果不拒絕虛無假設(在debate的前後其人們的想法並無改變)・也就是無統計上顯著的呈現在debate的前後人們有什麼想法上的改變。